Demo and Results Handbook

Statistical Characterization of a Gm-C Continuous-Time Filter in SKY130: Monte Carlo and PVT-Corner Study

Part	What it gives you
A	Showcase of every circuit: what to point at and what to say
B	Showcase of the Streamlit website: setup, tab-by-tab script, expected numbers
C	Every result in tables, each with a plain explanation
D	Division of labour for the four team members, demo run-of-show and Q&A owners
E	Appendix: all 45 PVT conditions for the three designs

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Spec used throughout: cutoff f_3 = 1.6 MHz ±10 % and Q between 0.6 and 0.8. A chip passes only if both hold. **Yield** = fraction of simulated chips that pass.

PART A

Showcase of every circuit

Five circuit-level views cover the whole design. Show them in this order: it goes from the big picture down to the transistors, then to how the circuits are fed, then to how they were tested.

#	Circuit / view	Show in	Time
1	Gm-C biquad (3 OTAs, 2 capacitors)	Figure A1	1 min
2	The OTA cell (5 transistors)	Figure A2	1 min
3	Transistor-level beta-multiplier current reference	Figure A3	1.5 min
4	The three bias schemes side by side	Figure A4	1 min
5	The test bench / Monte Carlo loop	Figure A5	1 min

Circuit 1 — The Gm-C biquad

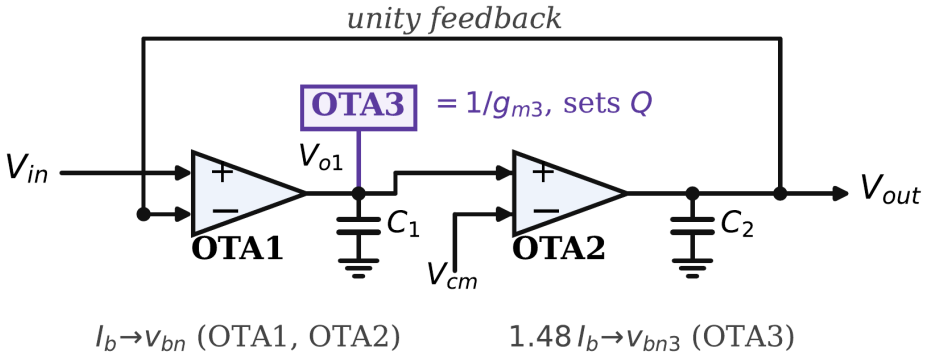


Figure A1 — Three-OTA, two-capacitor low-pass biquad.

Point at	Say
OTA1 and C1	“OTA1 is a tap: it turns the voltage difference between input and output into a current that charges C1.”
OTA2 and C2	“The second integrator. Two integrators give a second-order, steeper roll-off.”
Feedback wire to OTA1	“Output goes back to the input, so the DC gain is exactly 1.”
OTA3 labelled 1/gm₃	“OTA3 is wired as a resistor. It damps the filter and sets Q.”
Equations	ω₀ = gm/C and Q = gm/gm₃. With C = 10 pF, gm ≈ 100 µS gives 1.6 MHz. OTA3 is biased at 1.48× → Q ≈ 0.72.

Key numbers
f₃ = 1.610 MHz and Q = 0.722 at tt, 1.8 V, 27 °C. C1 = C2 = 10 pF. No common-mode feedback, no output buffer (a stated limitation).

Circuit 2 — The OTA cell

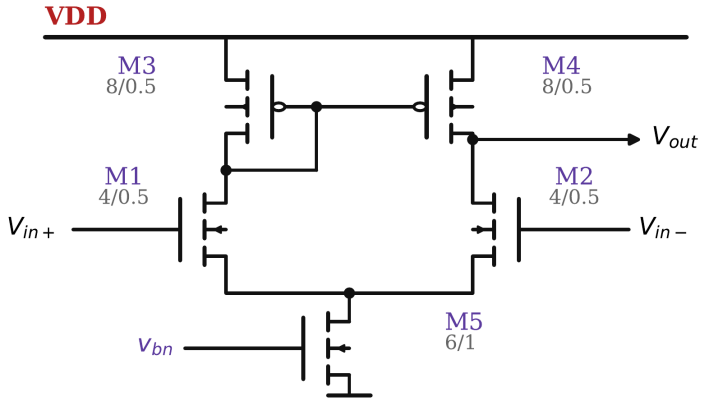


Figure A2 — The five-transistor OTA used for OTA1, OTA2 and OTA3.

Transistors	Size W/L (µm)	Role to say aloud
M1, M2 (NMOS pair)	4 / 0.5	Input pair. Wide for gm, short for low output capacitance.
M3, M4 (PMOS mirror)	8 / 0.5	Load. Twice as wide to put the output near mid-supply.
M5 (NMOS tail)	6 / 1	The tuning knob. Its gate voltage (vbn) sets the current, hence gm. Longer for a steadier current.

What to stress: the three OTAs are identical; only the tail bias differs (OTA3 gets 1.48× the current). All devices are 1.8 V `sky130_fd_pr` types.

Circuit 3 — The beta-multiplier current reference (the “real core”)

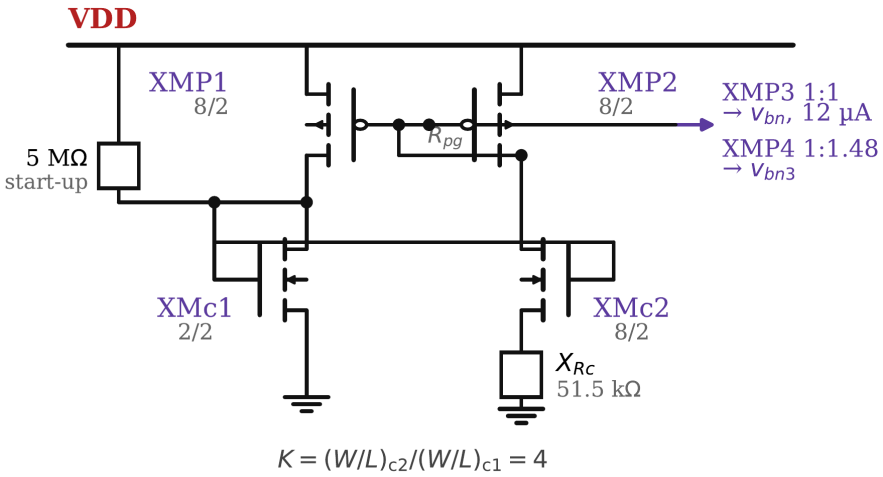


Figure A3 — Transistor-level core that replaces the ideal PTAT source.

Point at	Say
XMP1 / XMP2 (PMOS mirror, 8/2)	“A mirror forces equal current into two NMOS branches.”
XMc1 (2/2, diode-connected) and XMc2 (8/2) with the resistor	“One branch has a 4× wider transistor plus a ≈ 51.5 kΩ poly resistor. That imbalance fixes the current.”
Formula	$I \approx 2 / (\mu_n C_{ox}(W/L)\cdot R^2) \cdot (1 - 1/\sqrt{K})^2$, K = 4. Depends on resistor and mobility, to first order not on the supply.
XMP3 (1:1) and XMP4 (1:1.48)	“Two mirrors copy it to vbn for OTA1/2 and vbn3 for OTA3.”
5 MΩ start-up path	“Removes the zero-current state so the circuit always starts.”

Key numbers
Calibrated to 12.0 µA at tt, 1.8 V, 27 °C. Measured exponent 0.64–0.67 (we tuned for 0.6). Current 11.3 / 12.0 / 12.3 µA at 1.62 / 1.8 / 1.98 V, i.e. ≈ 0.5 % per 1 % of supply. That supply sensitivity is why the real core scores 71.7 % instead of the ideal 88.7 %.

Circuit 4 — Three bias schemes

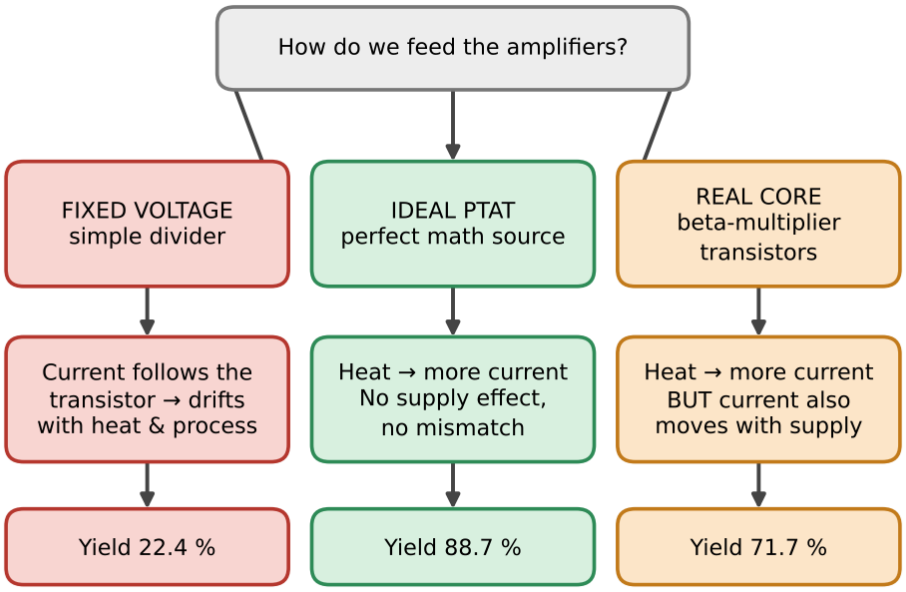


Figure A4 — The same filter fed three ways.

Say: “Everything else is identical. We only change how the tail current is made: a fixed voltage, an ideal source whose current rises with temperature, and a real transistor circuit that does the same thing.”

Circuit 5 — The test bench (how the circuits were exercised)

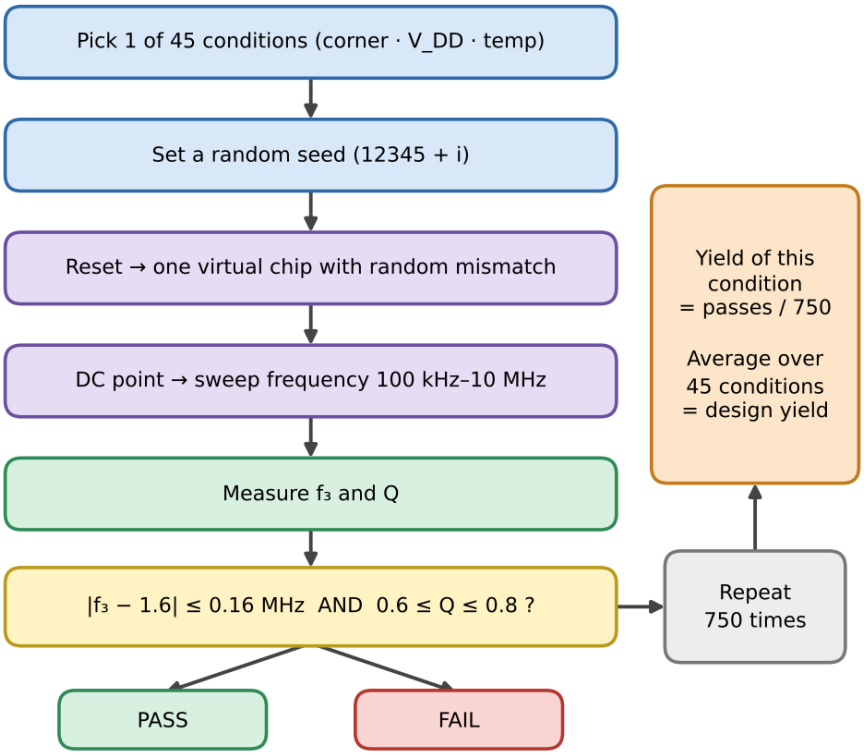


Figure A5 — One condition: seed, reset, solve, sweep, measure, pass/fail; repeat 750 times.

Item	Detail
Conditions	5 corners × 3 supplies × 3 temperatures = 45
Trials	750 per condition for voltage bias and ideal PTAT (33,750 each); 40 per condition for the real core (1,800)
Per trial	Reset → operating point → AC sweep 100 kHz–10 MHz (20 pts/decade) → measure f_3, f_{90}, peak, power
Seeds	12345 + i, fully reproducible
Simulators	ngspice 47 (main), Xyce 7.10 (independent check), SKY130A models, `_mm` mismatch corners

PART B

Showcase of the Streamlit website

The “live Monte Carlo and PVT lab” is a Streamlit app (`gmc_app.py`) with eight tabs. It drives the same netlists through ngspice in the tool container and plots results as they arrive.

Setup

```
# safe mode: replays saved data, no Docker needed
cd ~/projects/gmc-filter && GMC_FAKE=1 streamlit run gmc_app.py

# real mode: needs the Docker image (docker pull hpretl/iic-osic-tools)
cd ~/projects/gmc-filter && chmod +x run_live2.sh && streamlit run gmc_app.py

# if localhost does not open from WSL, use the Network URL it prints, or:
streamlit run gmc_app.py --server.port 8510

# clean old live runs before the demo (keeps the saved study data)
rm -f results_live_* curves_live_* mclive_* netlive_* loglive_* live_err.txt
```

Be honest about the mode
In safe mode the live tab replays recorded runs. Say: “This replays the recorded campaign; the live version runs the same netlists through ngspice.”

Tab-by-tab script (about 5–6 minutes)

Tab	What it shows	Click	Say / expected numbers
Spec (sidebar)	Target f_3 1.60 MHz, tolerance ±10 %, Q window 0.60–0.80	Leave as is	“This is the pass rule every yield number uses.”
1 Live Monte Carlo	Choose design, corner, VDD, temperature, runs, seed. Magnitude, cutoff zoom and phase update as runs finish; KPI tiles show f_3, Q, power, ripple, yield.	PTAT, tt, 1.8 V, 27 °C, runs 50 → **Run**. Then Voltage bias at −20 and 85 °C.	PTAT tt/27: f_3 ≈ 1.60 MHz, Q ≈ 0.72, σ(f_3) ≈ 5.7 %, yield ≈ 92 %. Blue lines = individual chips, yellow = mean ±3σ. Voltage bias moves to ≈ 1.2 MHz at −20 °C and ≈ 1.9 MHz at 85 °C.
2 PVT grid	Heat maps over corner × supply × temperature for the chosen dataset	Dataset: PTAT, then Voltage bias	Voltage bias is almost all dark; PTAT is mostly bright. Weakest PTAT cell: sf, 1.62 V, −20 °C.
3 Compare sets	Overlay histograms and box plots of f_3, Q, power for any saved result sets	Pick two or three sets	“Same circuit, different corner: the whole cloud shifts but keeps its shape.”
4 Pelgrom	Simulated σ(ΔVgs) against the hand formula $\sqrt{2}\cdot A_{VT}/\sqrt{(WL)}$, A_{VT} = 3.356 mV·µm	Look at the plot (button re-runs 500 samples)	“Points sit on the line: bigger area, smaller mismatch.”
5 Transient	Time-domain step response	Optional	Step overshoot ≈ 4.76 % (ideal Butterworth 4.3 %); AC and transient agree to 0.028 dB.
6 Bias comparison	Voltage bias against PTAT: conditions compared, mean yields, worst f_3 shift, bar charts, yield by corner	Scroll the two bar charts	Yield 22.4 % → 88.7 % (+66.3 points). Worst f_3 shift 0.944 → 0.084 MHz.

Tab	What it shows	Click	Say / expected numbers
7 Study results	Extra Monte Carlo sets: 1,500-run PTAT sets, tight tolerance, mirror ×1/×4/×16 at 27 °C and −20 °C, plus analysis output files	Read the table	Mirror ×4 cuts σ(f_3) 5.91 → 4.81 %; ×16 adds nothing.
8 Cross-checks and limits	Check table, Xyce-vs-ngspice scatter on y = x, nominal AC response, noise, limits	Show the scatter	Pelgrom 5.62 vs 5.91 %; Xyce within 0.003 %. “This validates the simulators, not the PDK mismatch numbers.”

> **If something fails on stage**
> Press **Stop**, move to PVT grid / Bias comparison / Study results. Those tabs read saved files, so they cannot fail. If the app will not start, show the heat maps and histograms from this handbook.

PART C

All results, tabulated and explained

Each table is followed by what it means in plain words. Source: the 45-condition campaign (33,750 trials for voltage bias and ideal PTAT; 1,800 for the real core).

Result 1 — Headline comparison

Quantity	Fixed voltage	Ideal PTAT	Real core
Trials per condition	750	750	40
Mean yield	22.4 %	**88.7 %**	71.7 ± 2 %
Lowest-condition yield	0.0 %	72.3 %	17.5 %
Worst f_3 shift from 1.6 MHz	0.944 MHz	0.084 MHz	0.285 MHz
Mean $\sigma(f_3)$	5.32 %	6.07 %	6.75 %
Range of mean Q	0.598–0.806	0.697–0.737	0.690–0.752
Power	28.0–132.5 µW	72.0–113.0 µW	n/a
Yield at tt, 1.8 V, 27 °C	95.5 %	92.3 %	85.0 %

What it means. With a plain voltage bias, only about one chip in five passes over the full PVT range. An ideal temperature-tracking current rescues almost 9 in 10; a real circuit that does the same still delivers about 7 in 10. Notice the fixed-voltage design is *better* at the one point it was tuned (95.5 % vs 92.3 %) and collapses everywhere else: single-point simulation hides the problem. The mismatch spread $\sigma(f_3)$ does not improve with PTAT (5.32 → 6.07 %): the reference moves where the cutoff sits, not how much it wobbles.

Result 2 — Where the yield is lost (breakdown by corner, supply, temperature)

Group	Fixed voltage	Ideal PTAT	Real core
corner = tt	31.5 %	88.8 %	80.0 %
corner = ss	32.7 %	89.1 %	70.6 %
corner = ff	29.0 %	88.3 %	68.6 %
corner = sf	6.5 %	87.5 %	65.0 %
corner = fs	12.4 %	89.9 %	74.2 %
VDD (V) = 1.62	22.8 %	84.3 %	64.0 %
VDD (V) = 1.8	22.3 %	91.0 %	76.2 %
VDD (V) = 1.98	22.2 %	90.9 %	74.8 %
T (°C) = -20	21.2 %	82.0 %	65.5 %
T (°C) = 27	19.2 %	89.5 %	72.0 %
T (°C) = 85	26.9 %	94.6 %	77.5 %

What it means. Fixed voltage: supply barely matters (22.8 / 22.3 / 22.2 %); the loss is process corner and temperature, with the mixed corners sf (6.5 %) and fs (12.4 %) worst. Ideal PTAT: roughly flat across corners (87.5–89.9 %), slightly weaker at 1.62 V (84.3 %) and cold (82.0 %). Real core: the signature is supply, 64.0 / 76.2 / 74.8 %, a low-supply dip the ideal source does not have. That is the core's own supply sensitivity.

Result 3 — Best and worst conditions

Design	Three worst conditions	Three best conditions
Fixed voltage	tt 1.98 V 85 °C: 0.0 % ss 1.62 V -20 °C: 0.0 % ss 1.8 V -20 °C: 0.0 %	ss 1.62 V 85 °C: 98.3 % ss 1.8 V 85 °C: 96.9 % tt 1.8 V 27 °C: 95.5 %
Ideal PTAT	sf 1.62 V -20 °C: 72.3 % ss 1.62 V -20 °C: 74.3 % tt 1.62 V -20 °C: 74.7 %	ss 1.8 V 85 °C: 96.5 % ss 1.98 V 85 °C: 96.5 % fs 1.8 V 85 °C: 96.4 %
Real core	sf 1.62 V -20 °C: 17.5 % sf 1.62 V 27 °C: 30.0 % ff 1.62 V -20 °C: 40.0 %	ff 1.98 V 85 °C: 95.0 % fs 1.62 V 85 °C: 95.0 % sf 1.98 V -20 °C: 95.0 %

What it means. Fixed voltage hits 0 % in several cells (e.g. tt at 1.98 V and 85 °C) and 98 % in a few where corner and temperature accidentally cancel (ss at 85 °C). The ideal PTAT worst case is still 72 %. The real core's single worst cell is sf, 1.62 V, −20 °C at 17.5 %: slow-NMOS/fast-PMOS corner, lowest supply and cold all push the same way. With only 40 trials per cell, individual core cells carry about ±15 points of uncertainty; only the 45-cell mean (±2) is tight.

Result 4 — Mean cutoff range over the grid

Design	Lowest mean f_3	Highest mean f_3	Worst shift from 1.6 MHz
Fixed voltage	0.666 MHz	2.544 MHz	0.944 MHz (≈ 59 %)
Ideal PTAT	1.516 MHz	1.638 MHz	0.084 MHz (≈ 5 %)
Real core	1.315 MHz	1.791 MHz	0.285 MHz (≈ 18 %)

What it means. This is the pure drift, with mismatch averaged out. The fixed-voltage cutoff roams from 0.67 to 2.54 MHz, a factor of almost 4. At tt and 1.8 V alone it goes from 1.20 MHz at −20 °C to 1.94 MHz at 85 °C. The ideal PTAT keeps every condition's mean inside 1.516–1.638 MHz.

Result 5 — The PTAT exponent scan

Exponent α	f_3 at −20 °C	f_3 at 27 °C	f_3 at 85 °C	Drift vs 27 °C
0.6 (chosen)	1.5935 MHz	1.6101 MHz	1.5970 MHz	−1.0 % / −0.8 %
1.0 (pure PTAT)	1.5035 MHz	1.6101 MHz	1.6937 MHz	−6.6 % / +5.2 %

What it means. A current that rises exactly in proportion to temperature (α = 1) overcorrects. Scaling it as $T^{0.6}$ cancels the mobility loss almost perfectly (within about 1 %). The real core's measured exponent, 0.64–0.67, lands close to this.

Result 6 — The transistor-level core, measured

Property	Value	Meaning
Temperature exponent	0.64 – 0.67	Measured from −40 to 125 °C at three corners and three supplies; close to the tuned 0.6.
Current at 1.62 / 1.8 / 1.98 V	11.3 / 12.0 / 12.3 µA	About 0.5 % per 1 % of supply: the core's weakness.
Calibration	12.0 µA at tt, 1.8 V, 27 °C	Resistor ≈ 51.5 kΩ (res_high_po_1p41); K = 4; start-up 5 MΩ.
Longer core devices (constant W/L)	no improvement	Area does not fix supply sensitivity; keep L = 2 µm.

Property	Value	Meaning
Mean f_3 vs supply	1.49 / 1.61 / 1.65 MHz	The supply shift carries straight into the cutoff.

What it means. The real reference reproduces the temperature law we wanted, so temperature drift is solved. What it cannot avoid is that its own current leans on the supply. That, plus the slightly steep exponent at cold, is the whole 17-point gap to the ideal source.

Result 7 — Mismatch distribution (Monte Carlo)

Case (ideal PTAT)	Trials	Mean f_3	σ(f_3)	Mean Q	σ(Q)	Yield
tt, 1.8 V, 27 °C	1,500	1.602 MHz	5.64 %	0.720	0.026	92.7 %
Worst PTAT condition	1,500	1.515 MHz	7.79 %	—	—	73.0 %

What it means. Random mismatch alone gives an approximately Gaussian cloud with about 5–6 % spread in cutoff and 0.026 in Q. At the typical condition 92.7 % of chips pass. At the worst corner the whole cloud shifts and widens, so only 73 % pass. f_3 and Q move together (a draw that raises current raises both), so failing trials tend to fail on both.

Result 8 — Does more area help? (mirror scaling)

Mirror	σ(f_3)	Yield effect	Power
×1	5.91 %	baseline	90.9 µW
×4	4.81 %	+5 points (27 °C), +6.6 points (−20 °C)	155.7 µW (+70 %)
×16	no further gain	none	≈ 4.6× baseline

Variance share of f_3	OTA load	Reference	Tail	Input pair
Mirror ×1	47 %	33 %	19 %	1 %
Mirror ×4	62 %	11 %	27 % (tail + input pair)	

What it means. Pelgrom says 4× area halves device mismatch, and it helps: σ drops from 5.91 to 4.81 % for 70 % more power. But at ×4 the reference contributes only 11 % of the variance and the OTA's own load devices dominate (62 %), so ×16 buys nothing. Diminishing returns, plus a power bill.

Result 9 — Cross-validation

Check	Result	Why it matters
Pelgrom sensitivity calculation	σ(f_3) 5.62 % vs 5.91 % (MC); ×4: 4.87 vs 4.81 %; σ(Q) 3.61 vs 3.68 %	An independent hand route reproduces the simulated spread.
AC vs transient	≤ 0.028 dB and ≤ 0.14°	The frequency-domain measurement is trustworthy.
Tighter solver tolerance	f_3 shift 0.12 % (max 0.18 %); Q max 0.41 %; yield unchanged 92.7 %	Numerical settings do not change the result.
Seed reproducibility	92.3 ± 1.9 % (750) and 92.7 ± 1.3 % (1,500); pooled 92.6 ± 1.1 %	Different random draws give the same answer within error.
Xyce 7.10, nominal	f_3 and Q within 0.001 %; operating point to ~6 digits	A second simulator agrees.
Xyce with injected mismatch (30 draws, 17 devices)	Per-run f_3 within 0.003 %, correlation 1.00000, σ = 5.23 % in both	Xyce ignores the PDK mismatch model, so offsets were injected: this validates the simulators, **not** the PDK numbers.

Result 10 — Secondary characteristics

Measurement	Value	Meaning
Output noise	240 µV rms (1 kHz–10 MHz)	About 50 dB SNR for a 100 mV peak input; high 1/f part (flicker coefficient flagged in the PDK).
THD	0.33 ± 0.06 % at 200 kHz, 100 mV	Low distortion over 200 mismatch trials.
1 dB compression	65 mV peak at 1.6 MHz	Usable input range.
Step overshoot	4.76 ± 1.08 %	Ideal Butterworth is 4.3 %: Q is slightly above Butterworth, as designed.
Passband peaking	0.033 dB typical; 0.350 dB largest of 33,750 PTAT trials	Negligible, so ripple is reported but not part of the pass rule.
Nominal response	f_3 = 1.610 MHz, Q = 0.722	Within 0.6 % of target; −90° point near 1.58 MHz.

Result 11 — Sensitivity to reference errors and statistical confidence

Item	Value	Meaning
5 % error in I_ref	costs 5–12 yield points	The reference current must be accurate to a few percent.
5 % error in the 1.48 ratio	costs 8–10 yield points	OTA1/2 and OTA3 legs must match well.
Yield uncertainty, 750 trials	±3.6 points per condition; ±0.4 on the 45-condition mean	Heat-map cells of the two 750-run designs are reliable to a few points.
Yield uncertainty, 40 trials	±15 points per cell; ±2 on the mean	Use only the mean for the real core.

PART D

Division of labour for four people

Jaidev carried the core technical work; the other three members hold defined support roles. It covers the work packages, who speaks in the demo and who answers which question.

Work packages

Owner	Role	Responsible for
Jaidev K (project lead, main contributor)	Technical lead	Core technical work: the Gm-C biquad and OTA design; the three bias schemes (fixed voltage, ideal PTAT, transistor-level beta-multiplier); the Monte Carlo and PVT campaign and its Python driver; the cross-checks (Pelgrom, Xyce, tolerance, seeds); the Streamlit lab; the results analysis, report, deck and IEEE paper.
S Jai Akaash	Literature and documentation support	Literature review and references; checking the report and deck text against the sources; collecting figures and captions.
Mohanakrishnan S	Simulation and results support	Running and collating batches of PVT conditions; checking result tables against the data files; preparing the heat-map and histogram figures.
Jeswant P M	Testing and demo support	Dry-running the Streamlit lab and the demo; keeping fallback screenshots; proofreading the paper and handbook; setup checklist on demo day.

> **Note**
> This is a suggested split. Edit the support roles to match what each person actually did. Whatever you write here, every member should be able to explain the part they are named on, because the evaluators can ask any of you.

Demo run-of-show (about 10 minutes)

Min	Segment	Speaker	Shows
0:00	Problem and objective: “how many chips work?”	Jaidev	Spec; chain of causes
1:00	Literature gap: what is new	S Jai Akaash	Related-work slide: open PDK vs filter-level statistics
2:00	Circuits 1–4: biquad, OTA, bias schemes, beta-multiplier	Jaidev	Figures A1–A4; nominal run; drift of voltage bias; PTAT fix
5:30	Monte Carlo and the lab	Jaidev, with Jeswant driving the screen	Live Monte Carlo, PVT grid, Pelgrom tab
8:00	Results table and cross-checks	Mohanakrishnan	Bias comparison, Study results; reads the headline numbers
9:15	Conclusion, limits, next step	Jaidev	22.4 / 88.7 / 71.7 %; regulated core as next step

Who answers which question

Likely question	Answerer
Design questions: three OTAs, Q, PTAT, exponent 0.6, beta-multiplier, why the core loses 17 points	Jaidev
Monte Carlo method, number of trials, seeds, mismatch model validity, Pelgrom, Xyce	Jaidev

Likely question	Answerer
What is the related work gap? Which papers did you build on?	S Jai Akaash
Where do the numbers in the results table come from? How was yield counted?	Mohanakrishnan
How do you run the lab? What does each tab show?	Jeswant
What would you do next (regulated core, CMFB, layout)?	Jaidev

Before the demo checklist

Task	Owner
Start the app (safe mode), clean old live runs, keep the terminal open	Jeswant
Open figures A1–A5 (this handbook) on a second screen	Mohanakrishnan
Have the PDF deck and report open as backup	S Jai Akaash
Final run-through of numbers: 22.4 / 88.7 / 71.7 %	Jaidev

PART E

Appendix: all 45 conditions

Mean cutoff and yield per condition. Fixed voltage and ideal PTAT: 750 trials each; real core: 40 trials each (±15 points per cell).

Corner	VDD	T	f_3 volt.	Yield volt.	f_3 PTAT	Yield PTAT	f_3 core	Yield core
tt	1.62	-20	1.157	0.1	1.530	74.7	1.463	60.0
tt	1.62	27	1.578	94.5	1.559	86.3	1.492	72.5
tt	1.62	85	1.901	1.9	1.556	92.5	1.527	95.0
tt	1.8	-20	1.202	0.1	1.584	84.9	1.554	82.5
tt	1.8	27	1.622	95.5	1.603	92.3	1.607	85.0
tt	1.8	85	1.944	0.1	1.592	95.7	1.632	87.5
tt	1.98	-20	1.227	0.1	1.615	86.9	1.610	77.5
tt	1.98	27	1.647	91.2	1.629	90.4	1.646	82.5
tt	1.98	85	1.971	0.0	1.613	95.3	1.681	77.5
ss	1.62	-20	0.850	0.0	1.521	74.3	1.480	62.5
ss	1.62	27	1.251	0.7	1.550	85.5	1.572	87.5
ss	1.62	85	1.590	98.3	1.547	90.9	1.550	92.5
ss	1.8	-20	0.884	0.0	1.574	85.2	1.619	82.5
ss	1.8	27	1.285	1.3	1.593	92.7	1.704	65.0
ss	1.8	85	1.624	96.9	1.582	96.5	1.719	72.5
ss	1.98	-20	0.902	0.0	1.604	87.7	1.690	67.5
ss	1.98	27	1.304	2.0	1.618	92.5	1.745	65.0
ss	1.98	85	1.645	95.2	1.604	96.5	1.791	40.0
ff	1.62	-20	1.523	77.6	1.537	75.1	1.394	40.0
ff	1.62	27	1.943	2.0	1.567	86.4	1.432	52.5
ff	1.62	85	2.237	0.0	1.565	93.9	1.452	57.5
ff	1.8	-20	1.582	90.4	1.593	84.7	1.464	57.5
ff	1.8	27	1.999	0.1	1.613	91.2	1.534	72.5
ff	1.8	85	2.290	0.0	1.600	95.2	1.551	90.0
ff	1.98	-20	1.615	91.1	1.625	84.9	1.528	67.5
ff	1.98	27	2.031	0.0	1.638	88.1	1.632	85.0
ff	1.98	85	2.322	0.0	1.620	95.1	1.604	95.0
sf	1.62	-20	1.789	41.2	1.516	72.3	1.315	17.5
sf	1.62	27	2.180	0.0	1.551	83.9	1.394	30.0
sf	1.62	85	2.430	0.0	1.552	91.1	1.455	52.5
sf	1.8	-20	1.875	12.7	1.586	83.7	1.490	55.0
sf	1.8	27	2.258	0.0	1.605	91.6	1.543	82.5
sf	1.8	85	2.502	0.0	1.593	95.3	1.599	92.5
sf	1.98	-20	1.917	4.7	1.621	85.2	1.568	95.0

Corner	VDD	T	f_3 volt.	Yield volt.	f_3 PTAT	Yield PTAT	f_3 core	Yield core
sf	1.98	27	2.300	0.0	1.633	89.3	1.637	77.5
sf	1.98	85	2.544	0.0	1.616	95.3	1.662	82.5
fs	1.62	-20	0.666	0.0	1.534	75.7	1.484	57.5
fs	1.62	27	1.046	0.0	1.562	88.3	1.570	87.5
fs	1.62	85	1.397	25.1	1.558	93.5	1.575	95.0
fs	1.8	-20	0.691	0.0	1.581	86.5	1.599	80.0
fs	1.8	27	1.071	0.0	1.601	92.9	1.698	67.5
fs	1.8	85	1.423	38.0	1.591	96.4	1.701	70.0
fs	1.98	-20	0.703	0.0	1.609	88.0	1.643	80.0
fs	1.98	27	1.084	0.0	1.625	91.6	1.714	67.5
fs	1.98	85	1.438	48.3	1.610	96.0	1.750	62.5